Q(F) AGREES WITH F IN THE ETALE LOCAL RINGS OVER FLAT OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Allowed standard prerequisites (editorial).

- delivered companion: adjoint-kernel lemma. delivered companion lemma-adjoint-kernel-parasitic. Whole argument.

Source reference. Original lemma 0779: sources/stacks-chow/tags.txt, source lines 9194–9194. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Let x be an object of $\mathcal{X}$ lying over the scheme U such that $x : U \to \mathcal{X}$ is flat. Then for $\mathcal{F}$ in $QCoh^{fbc}(\mathcal{O}_\mathcal{X})$ we have $Q(\mathcal{F})|_{U_{étale}} = \mathcal{F}|_{U_{étale}}$.*

Proof. True because the kernel and cokernel of $Q(\mathcal{F}) \to \mathcal{F}$ are parasitic, see Lemma 0779. □

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